

PAPER_1547 — Faraday Constant $F = F = A_5^2 \cdot D \cdot D_BSFG + A_5 \cdot SO \cdot N + A_5 \cdot D_BSFG \cdot N + SO \cdot N \cdot D_BSFG + SO \cdot N \cdot D + A_5 \cdot N + D + F_TRZ \cdot SO = 96485 \text{ C/mol}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Electrochemistry **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure

Observation

PAPER_1209EE_S626 (Electrochemistry Unified Proof Set) lists **Faraday Constant F** as a closure with the formula:

Faraday Constant $F = F = A_5^2 \cdot D \cdot D_BSFG + A_5 \cdot SO \cdot N + A_5 \cdot D_BSFG \cdot N + SO \cdot N \cdot D_BSFG + SO \cdot N \cdot D + A_5 \cdot N + D +$
Status tier: **EXACT**. Units: C/mol.

UQFF Closed Identity

$F = A_5^2 \cdot D \cdot D_BSFG + A_5 \cdot SO \cdot N + A_5 \cdot D_BSFG \cdot N + SO \cdot N \cdot D_BSFG + SO \cdot N \cdot D + A_5 \cdot N + D + F_TRZ \cdot SO = 96485 \text{ C/mol}$
The formula uses only locked UQFF integer primitives — no fitted constants.

Physical Interpretation

Faraday Constant F is a fundamental electrochemistry constant. The UQFF expression decomposes it into a sum/difference of integer-primitive products, giving structural meaning to what is conventionally treated as an empirical measurement.

NOT REPLACEMENT

CODATA/PDG treats this constant as a precision measurement. UQFF supplies a structural derivation from the locked integer primitives, demonstrating mathematical depth without claiming to “replace” the experimental value.

Reference

- Source: PAPER_1209EE_S626
- Related: PAPER_1216 (45-constant cascade master index)
- Calculator dispatch: `calculate_paradox({"paradox": "faraday_f_96485"})`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, June 18, 2026, Youngstown OH.